History · Satyagraha and Non-Cooperation

Lesson

Courses · 148 days · A

Social Science · Class 9–10 > Unit 10: Gandhi and the National Mo... · Lessons

Satyagraha and the Non-Cooperation Movement

Gandhi's idea of satyagraha, his early struggles in India, the Rowlatt Act and Jallianwala Bagh, and the Khilafat and Non-Cooperation Movement (1920–1922).

18 min · 45 energy points · Ask Questy

Article · Practice · Flashcards

Bookmark · Notes · Add a doubt

Open notebook

Listen to this lesson



Gandhi returned to India from South Africa in January 1915 with a new way of fighting: satyagraha.

Mahatma Gandhi had fought racist laws in

Worked example

Reading different meanings

The peasants of Awadh, the tribals of Andhra and the plantation workers of Assam all joined Non-Cooperation. Did they understand 'swaraj' the same way as Gandhi?

1. Look at each group's own problem: rents (Awadh), forest laws (Andhra), freedom to move (Assam).
2. Notice their methods: some used force, which Gandhi did not approve.
3. Conclude: each group gave swaraj its own meaning, linked to its own suffering.

Answer: No. For each group, swaraj meant the end of its own particular hardship, and some used methods Gandhi rejected. The movement grew strong because many different hopes joined under one banner.

Common mistake

Common mistake

✗ **Wrong:** Gandhi withdrew the Non-Cooperation Movement because it was failing.

✓ **Right:** The movement was strong and growing. Gandhi withdrew it in 1922 because of the **violence at Chauri Chaura**; he believed satyagraha must stay non-violent.

Usually movements stop when they lose support, so students assume that happened here. Many leaders were actually unhappy with the withdrawal for exactly that reason.

Guided activity

Try it

Put these events in order.

1. Gandhi returns from South Africa
2. Champaran Satyagraha
3. Jallianwala Bagh massacre
4. Non-Cooperation adopted at Nagpur
5. Chauri Chaura incident

1915 → 1917 → April 1919 → December 1920 → February 1922.

Do it again

Geography · Farming types and crop seasons

Lesson

Courses · 148 days · A

Social Science · Class 9–10 › Unit 30: Land Use and Agriculture

Lessons

Types of Farming and Cropping Seasons

Subsistence, commercial and plantation farming, and the rabi, kharif and zaid seasons.

15 min · 40 energy points · Ask Questy

Article | Practice | Flashcards

Bookmark · Notes · Add a doubt

Open notebook

Listen to this lesson



A farmer in Mandya plants paddy when the June rains begin; a farmer in Punjab sows wheat in November. Different crops, different seasons.

India is an **agricultural** country: a large

Worked example

Explaining a farming type

A farmer in Bihar has only 1 hectare of land and a big family. He uses fertilisers and irrigation and grows two or three crops a year. Name the farming type and explain.

1. The land is small and the population is dense.
2. He uses heavy inputs (fertilisers, irrigation, labour) to get the most from the land.
3. The food is mainly for the family, so it is subsistence.

Answer: Intensive subsistence farming.

Common mistake

Common mistake

✗ **Wrong:** Rabi crops are harvested in winter, since they are 'winter crops'.

✓ **Right:** Rabi crops are **sown** in winter (October–December) and **harvested** in summer (April–June).

The label 'winter crop' makes students think of the harvest time. Rabi crops grow through the winter and ripen in the warm spring.

Guided activity

Try it

Sort each crop into its cropping season.

3 / 8

Cucumber

Rabi

Kharif

Zaid

✓ Right!

Rabi

Kharif
Jowar

Zaid
Watermelon

Economics · Consumer rights

Lesson

Courses · 148 days · A

Social Science · Class 9–10 › Unit 60: Consumer Rights and Cons... · Lessons

Consumer Exploitation, Rights and Duties

Ways consumers are cheated, the consumer movement in India, the six consumer rights, quality marks and the duties of a consumer.

16 min · 40 energy points · Ask Questy

Article · Practice · Flashcards

Bookmark · Notes · Add a doubt

Open notebook

Listen to this lesson



A packet of chips says 100 g, but weighs 85 g. A shopkeeper charges more than the MRP for a cold drink. These are everyday cases of consumer exploitation.

Worked example

Reasoning: spotting the right violated

Sana buys a ₹40 cold drink with an MRP of ₹35. The shopkeeper says "₹5 extra for cooling". When she refuses, he tells her she must also buy chips if she wants the drink. Which rights are violated, and what should she do?

1. Charging ₹40 when MRP is ₹35 is **overcharging**, an unfair trade practice. MRP is the maximum, including all taxes, so no extra can be charged.
2. Forcing her to buy chips with the drink violates the **right to choose**.
3. Her **duty** is to ask for a proper bill and not pay more than MRP.
4. If needed, she can complain on the **National Consumer Helpline** or file a complaint with the **District Commission** (right to seek redressal).

Answer: Overcharging above MRP and the right to choose are violated. She should insist on MRP, take a bill, and complain if the seller refuses.

Common mistake

Common mistake

✗ **Wrong:** A shop can charge more than the MRP if it keeps the drink cold or adds tax.

✓ **Right: MRP is the maximum retail price including all taxes. No one can legally charge more than MRP; one can bargain for less.**

Shops often add "cooling charges" and people pay without thinking. The word "maximum" is the key: MRP is a ceiling, not a starting price.

Guided activity

Try it

Match each situation to the consumer right involved.

3 / 5

A company refuses to replace a faulty fridge

- Right to seek redressal
- Right to safety
- Right to be informed

✓ **Right!**

Consumer groups are invited to give views on a new rule

→ Right to represent

A dealer insists you buy his brand of battery

→ Right to choose

Maths · Constructing tangents (Karnataka SSLC chapter)

Lesson

Courses · 148 days · A

High School Maths › Unit 23: Constructions · Lessons

Constructing Tangents to a Circle

Draw the tangent at a point on a circle, the pair of tangents from an outside point, and a pair of tangents at a given angle, with the reason each works.

17 min · 45 energy points · Ask Questy

Article · Practice · Flashcards

Bookmark · Notes · Add a doubt

Open notebook

Listen to this lesson

Wrap a rope round a well and pull both ends to one point P outside it. The two straight parts just touch the well's rim. They are **tangents** from P. In this lesson you draw them exactly with a ruler and compasses, instead of placing the ruler "by eye".

Two facts from the chapter on circles make every construction here work:

Worked example

Tangents and their length

Draw a circle of radius 6 cm. From a point P 10 cm from its centre, construct the pair of tangents and find their length.

1. Draw the circle with centre O, radius 6 cm. Mark P with OP = 10 cm.
2. Bisect OP to get M (MO = 5 cm). With centre M and radius 5 cm draw a circle cutting the first circle at Q and R.
3. Join PQ and PR: these are the tangents.
4. Check by calculation: in right △OQP, $PQ = \sqrt{OP^2 - OQ^2} = \sqrt{100 - 36} = \sqrt{64}$.

Answer: PQ = PR = 8 cm (measure to confirm).

Common mistake

Common mistake

✗ **Wrong:** For tangents inclined at 60°, I draw two radii with 60° between them and draw perpendiculars at their ends.

✓ **Right: Make the angle between the radii 180° – 60° = 120°. Then the tangents meet at 60°.**

It feels natural to copy the given angle. But in OAPB two angles are already 90°, so the angle at O and the angle at P must add up to 180°. Radii at 60° would give tangents at 120°.

Guided activity

Try it

Put the steps for the tangents from P in order.

1. Join P to the centre O.
2. Draw the perpendicular bisector of PO to find its mid-point M.
3. With centre M and radius MO, draw a circle. It cuts the given circle at Q and R.
4. Join PQ and PR. These are the two tangents.

The new circle has **PO as a diameter**. That is the whole trick: it makes the angle at Q a right angle.

Do it again

Science · Sources of energy (Karnataka SSLC chapter)

Lesson

Courses · 148 days · A

School Physics › Unit 12: Sources of Energy · Lessons

Conventional Sources of Energy

What makes a good fuel, fossil fuels, thermal and hydro power plants, biomass and biogas, and wind energy.

17 min · 45 energy points · Ask Questy

Article · Practice · Flashcards

Bookmark · Notes · Add a doubt

Open notebook

Listen to this lesson



Your mother cooks on an LPG stove, the fan runs on electricity from a power plant, and the bus to school burns diesel. Each one uses a different source of energy.

Worked example

Power from a dam

At a dam, 1000 kg of water falls through a height of 50 m every second. How much power is available? (Take $g = 10\ \text{m/s}^2$.)

1. Potential energy lost by the water each second $= mgh$.
2. $mgh = 1000 \times 10 \times 50 = 5 \times 10^5$ J.
3. Power $= \frac{\text{energy}}{\text{time}} = \frac{5 \times 10^5\ \text{J}}{1\ \text{s}} = 5 \times 10^5$ W.
4. A real plant turns only part of this into electricity, because some energy is lost in the turbine and generator.

Answer: 5×10^5 W, that is 500 kW, at most.

Common mistake

Common mistake

✗ **Wrong:** Biogas is made by burning cow dung.

✓ **Right:** Biogas is made when bacteria **decompose** cow dung and water **without oxygen** (anaerobic decomposition) in a closed digester.

Cow-dung cakes are burnt as fuel in villages, so it seems biogas must come from burning too. But burning would use up the dung and give smoke and ash; in a biogas plant nothing burns until the gas itself reaches the stove.

Guided activity

Try it

Put the steps of a thermal power plant in order.

1. Coal is burnt in a furnace
2. Heat boils water into high-pressure steam
3. Steam turns the blades of a turbine
4. The turbine shaft turns the generator
5. Electricity is sent out on transmission lines

Energy changes step by step: **chemical** energy of coal → **heat** → **kinetic** energy of steam and turbine → **electrical** energy.

Do it again

English · Reported speech

Lesson

Courses · 148 days · A

English > Unit 15: Reported Speech · Lessons

Reporting Statements

Direct and indirect speech, said/told, tense back-shift, pronoun changes, time and place words, and when the tense does not change.

16 min · 42 energy points · Ask Questy

Article · Practice · Flashcards

Bookmark · Notes · Add a doubt

Open notebook

Listen to this lesson



Anil said, "I am going to Hampi tomorrow." A day later his friend tells you: "Anil said that he was going to Hampi the next day."

In **direct speech** we repeat the speaker's exact words inside **inverted commas**

Worked example

Changing a statement step by step

Change into indirect speech: Fatima said to me, "I bought this saree in Mysuru yesterday."

1. Reporting verb: *said to me* → **told me**; add **that**.
2. Pronoun: *I* (Fatima) → **she**.
3. Tense: simple past *bought* → past perfect **had bought**.
4. Words: *this* → **that**; *yesterday* → **the day before** (or *the previous day*).
5. Remove the inverted commas and the comma after *said*.

Answer: Fatima **told me that she had bought that saree** in Mysuru **the day before**.

Common mistake

Common mistake

✗ **Wrong:** She said me that she was busy.

✓ **Right:** She **told me** that she was busy. (or: She **said that** she was busy.)

In Indian English *said me* is heard often, so it feels normal. But *say* does not take a person directly as object; use **said to me** in direct speech and **told me** in indirect speech.

Guided activity

Try it

Match each direct word with its reported form.

3 / 6

this

- **the day before**
- **then**
- **before**
- **that**

✓ **Right!**

here → there

tomorrow → the next day

Kannada · Sandhi (lopa, agama, adesha)

Lesson

Courses · 148 days · A

Kannada · Grammar and Writing
Unit 1: Sandhi (ಸಂಧಿ)

Lessons

Kannada Sandhi: ಲೋಪ, ಆಗಮ, ಆದೇಶ

What sandhi is, and the three Kannada sandhis: a vowel drops (ಲೋಪ), a letter is added (ಆಗಮ), or a letter is replaced (ಆದೇಶ).

15 min · 40 energy points · Ask Questy

Article · Practice · Flashcards

Bookmark · Notes · Add a doubt

Open notebook

Listen to this lesson

When we speak fast, two words stick together: **ಊರು + ಅಲ್ಲಿ** becomes **ಊರಲ್ಲಿ** (ooralli — in the town).

When two sounds meet and join without a gap, the joining is called **ಸಂಧಿ** (sandhi —

Worked example

Name the sandhi

Split ಮನೆಯಲ್ಲಿ (maneyalli — in the house) and name the sandhi.

1. Find the two parts: ಮನೆ (house) + ಅಲ್ಲಿ (in).
2. A vowel meets a vowel: ಮನೆ ends in ಎ, and ಅಲ್ಲಿ starts with ಅ.
3. Try dropping the ಎ: we get ಮನಲ್ಲಿ, which has no meaning. So ಲೋಪ is not possible.
4. A new letter ಯ has come in: ಮನೆ + ಯ್ + ಅಲ್ಲಿ = ಮನೆಯಲ್ಲಿ.

Answer: ಮನೆ + ಅಲ್ಲಿ = ಮನೆಯಲ್ಲಿ — **ಯಕಾರಾಗಮ ಸಂಧಿ** (ಆಗಮ ಸಂಧಿ).

Common mistake

Common mistake

✗ **Wrong:** ಗುರುವನ್ನು is ಲೋಪ ಸಂಧಿ, because ಗುರು ends in ಉ just like ಊರು.

✓ **Right: ಗುರುವನ್ನು is ವಕಾರಾಗಮ ಸಂಧಿ: ಗುರು + ಅನ್ನು = ಗುರುವನ್ನು.**

Both words end in ಉ, so it feels the same. But if the ಉ of ಗುರು dropped, we would get ಗುರನ್ನು, which is wrong. A ವ is added, so it is ಆಗಮ. Always check: did a letter drop, or did a new letter come in?

Guided activity

Try it

Which Kannada sandhi is each word?

3 / 6

ಊರಲ್ಲಿ (ಊರು + ಅಲ್ಲಿ)

- ಲೋಪ ಸಂಧಿ
- ಆಗಮ ಸಂಧಿ
- ಆದೇಶ ಸಂಧಿ

✓ Right!

ಲೋಪ ಸಂಧಿ

ಆಗಮ ಸಂಧಿ
ಕೈಯನ್ನು (ಕೈ + ಅನ್ನು)

ಆದೇಶ ಸಂಧಿ
ಮಳೆಗಾಲ (ಮಳೆ + ಕಾಲ)

Hindi · Vachya (voice)

Lesson

Courses · 148 days · A

Hindi · Grammar and Writing › Unit 3: Voice (वाच्य)

Lessons

कर्तृवाच्य, कर्मवाच्य और भाववाच्य

The three voices of Hindi, how the verb follows the doer, the object or neither, and step-by-step voice change.

17 min · 42 energy points · Ask Questy

Article · Practice · Flashcards

Bookmark · Notes · Add a doubt

Open notebook

Listen to this lesson

🎯

Who is the hero of the sentence: the doer, the thing done, or just the action? The answer gives you the वाच्य.

वाच्य (voice) is the form of the verb that shows what is most important in the

Worked example

कर्तृवाच्य → कर्मवाच्य

Change into कर्मवाच्य: 'माँ ने रोटियाँ बनाईं। (Mother made rotis.)'

1. Find the parts: कर्ता 'माँ', कर्म 'रोटियाँ' (स्त्रीलिंग, बहुवचन), verb 'बनाईं' (भूतकाल).
2. Remove 'ने' and put 'द्वारा' (or 'से') after the कर्ता: 'माँ द्वारा'.
3. Put the verb in its past form and add 'जाना' in the same tense: बनाई + गईं.
4. Make the verb agree with the कर्म 'रोटियाँ' (स्त्रीलिंग, बहुवचन): 'बनाई गईं'.

Answer: माँ द्वारा रोटियाँ बनाई गईं। (Rotis were made by mother.)

Common mistake

Common mistake

✗ **Wrong:** 'सीमा ने कविता लिखी' is कर्मवाच्य, because the verb 'लिखी' agrees with 'कविता' (the कर्म).

✓ **Right: It is कर्तृवाच्य. With 'ने' in the past tense, the verb follows the कर्म, but the sentence is still active.**

In Hindi, 'ने' makes the verb agree with the object, which looks like कर्मवाच्य. The real sign of कर्मवाच्य is 'से / द्वारा' after the कर्ता and the 'जाना' form of the verb (लिखी गई).

Guided activity

Try it

Which वाच्य is each sentence in?

3 / 6

किसान खेत जोतता है।

- कर्तृवाच्य
- कर्मवाच्य
- भाववाच्य

✓ Right!

कर्तृवाच्य

कर्मवाच्य
मुझसे यह पत्र नहीं लिखा गया।

भाववाच्य
आओ, अब सोया जाए।

Computer Applications · Arrays and searching (ICSE Java)

Lesson

Courses · 148 days · A

Computer Applications · Class 9...
Unit 12: Arrays
Lessons

Arrays, Linear Search and Binary Search

Declare, initialise and fill a single dimensional array, use length and indexes, and search it with linear search and binary search.

18 min · 45 energy points · Ask Questy

Article · Practice · Flashcards

Bookmark · Notes · Add a doubt

Open notebook

Listen to this lesson

To store the marks of 40 students you would need 40 variables. An **array** solves this: it is a group of variables of the **same type**, under **one name**, stored together. Each value is an **element**, reached by its **index** (subscript) in square brackets.

Java — Ways to create a single dimensional array

```
int marks[] = new int[5];          // 5 in
```

Worked example

Dry run of binary search

Trace BinarySearch for key = 41, then for key = 20. Give the output each time.

1. Key 41: low = 0, high = 6 → mid = 3, a[3] = 27 < 41, so low = 4.
2. low = 4, high = 6 → mid = 5, a[5] = 41 = key, so pos = 5 and the loop breaks.
3. Key 20: mid = 3 (27 > 20) → high = 2; mid = 1 (12 < 20) → low = 2; mid = 2 (19 < 20) → low = 3.
4. Now low = 3 > high = 2, so the loop ends with pos = −1.

Answer: Key 41:
Enter number to search: 41
Found at index 5

Key 20:
Enter number to search: 20
Not found

Common mistake

Common mistake

✗ **Wrong:** Binary search can be used on any array; it is just faster.

✓ **Right:** Binary search works **only on a sorted array**. On an unsorted array it may say "Not found" even when the value is there. Use linear search, or sort first.

The method throws away half the array after one comparison. That is safe only if every smaller value is on the left and every bigger value is on the right, which is true only when the array is sorted.

Guided activity

Try it

Order the steps of one round of binary search.

1. Find mid = (low + high) / 2
2. Compare a[mid] with the key
3. If equal, record the position and stop
4. If a[mid] < key, set low = mid + 1
5. Otherwise set high = mid − 1 and repeat while low <= high

Each round finds the middle, compares, and either stops or discards half of the array by moving low or high.

Do it again

GK · India's space journey

Lesson

Courses · 148 days · A

General Knowledge › Unit 6: Space and ISRO

Lessons

India's Space Journey

Thumba, Aryabhata, Rakesh Sharma, Chandrayaan, Mangalyaan and Aditya-L1, plus why astronauts float.

16 min · 40 energy points · Ask Questy

Article | Practice | Flashcards

Bookmark | Notes | Add a doubt

Open notebook

Listen to this lesson



In 1963, parts of India's first rocket were carried to the launch pad on a bicycle, at a fishing village called Thumba in Kerala.

India's space story began with very little money but big dreams. Physicist **Vikram**

Worked example

Why launch from Sriharikota?

ISRO launches from Sriharikota on the east coast, not from, say, Delhi. Give the reasons.

1. The Earth spins from **west to east**. A rocket launched **eastwards** gets a free push from this spin.
2. The push is biggest near the **equator**, where the Earth's surface moves fastest. Sriharikota is fairly far south (about 13°N).
3. To the east lies the **Bay of Bengal**, so spent rocket stages fall into the sea, not on towns.
4. It is an island with few people, so it is safer.

Answer: Its east-coast, southern location gives an eastward boost from the Earth's spin and a safe sea path for falling rocket parts.

Common mistake

Common mistake

✗ **Wrong:** Astronauts float because there is no gravity in space.

✓ **Right: At the height of the International Space Station (about 400 km), gravity is still about 90% as strong as on the ground. Astronauts float because they and their station are falling around the Earth together (free fall), so nothing pushes up on them.**

On Earth, floating usually means no pull. But an orbit is a fall that keeps missing the ground, because the station moves sideways so fast. Without gravity, the station would fly off in a straight line.

Guided activity

Try it

Put India's space milestones in order, earliest first.

1. First rocket launched from Thumba
2. Aryabhata launched
3. Rakesh Sharma goes to space
4.
5.
6.

Tap what comes next:

- **Chandrayaan-3 lands near the lunar south pole**
- **Chandrayaan-1 reaches the Moon**
- **Mangalyaan enters Mars orbit**

✗ That comes later. Which one happens first?